

Builder Basics · Farm House — English Worksheet

Topic: Farm House (House + Adjacent Farm) · **Course:** Builder Basics · **Time:** about 45 minutes

1 · Predict 🧠

Read each set of steps. Write what you think will happen.

```
set size to 3
set height to 2

repeat height times:
  repeat 4 times:
    repeat size times:
      place block down
      move forward
    turn right
  move up by 1
```

What does the agent build? How tall is it?


```
set size to 3

repeat size times:
  repeat size times:
    place block down
    move forward
  turn right
  move forward
  turn right
  repeat size times:
    move forward
  turn left
  turn left
```

What shape does the agent leave on the ground? What is it for in a farm house?


```
move down by 2
turn right
repeat 4 times:
    move forward
turn left
```

The agent just finished a house. Where is it going next?

2 · Spot the Bug 🐛

Each code block is broken. Read what it should do, rewrite it, then explain why the original was wrong.

Bug A — The agent is supposed to build the **house walls first**, then the **farm patch beside it** (two clear stages).

```
# Stage 1: farm patch
repeat 3 times:
  repeat 3 times:
    place block down
    move forward
  turn right
# Stage 2: house walls
repeat 2 times:
  repeat 4 times:
    repeat 3 times:
      place block down
      move forward
    turn right
  move up by 1
```

Write the fixed code:

Why was the order wrong? Why does your fix work?

Bug B — The agent is supposed to walk to the spot beside the house, **then** build the farm patch. The farm should land on the ground, not on top of the house.

```
repeat 3 times:
  repeat 3 times:
    place block down
    move forward
  turn right
move down by 2
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The agent is supposed to use the **same** **size** **variable** for the house walls and the farm patch, so changing one number changes both.

```
repeat 4 times:
  repeat 3 times:
    place block down
    move forward
  turn right
# farm patch
repeat 5 times:
  repeat 5 times:
    place block down
    move forward
  turn right
```

Write the fixed code using one **size** variable:

Why was it wrong? Why does your fix work?

3 · Show Your Work 📷 🎥

Switch to your homework world and build your farm house — small house with walls and roof, plus a farm patch beside it. Try making the farm bigger (5×5) **or** add a path between the house and farm.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit 

Send this worksheet + your video to teacher on KakaoTalk.